

Kotlin Goodies for Testing

Pasha Finkelshteyn, BellSoft

layout: two-cols

whoami

-    developer
- Average  enjoyer
- Developer  at BellSoft
- Data engineer
- Speaker 

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layout: center

Who are you?

1. You write code/tests in Java or Kotlin
2. You wanna migrate to Kotlin, *or*
3. You wanna make your tests in Kotlin better, *or*
4. You wanna introduce Kotlin in your projects!

layout: center

How are we doing it?

1. Using DSLs (domain specific languages)
2. Structuring our code better
3. Using smart libraries

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BDD

1. Given
2. When
3. Then

clicks: 3

Cucumber

```
1  Feature: An example
2    Scenario: The example
3      Given I have a username "pasha"
4      When I type username in field "x"
5      And Press OK button
6      Then I get an error "Enter password"
```

What's the problem here?

clicks: 3

Cucumber

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1  Feature: An example
2    Scenario: The example
3      Given I have a username "pasha"
4      When I type username in field "x"
5      And Press OK button
6      Then I get an error "Enter password"
```

This line might intersect with **any** test

clicks: 3

Cucumber

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Actually **any** line can

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Actually **any** line can

Steps match by **name** O_o:

How can we do it in Kotlin?

Kotest: <https://kotest.io>

- No obscure text files
- No string matching
- everything in one place

```
3         `when`("I sit on it") {  
4             then("I should be able to fly") {  
5                 // test code  
6             }  
7         }  
8         `when`("I throw it away") {  
9             then("it should come back") {  
10                // test code  
11            }  
12        }  
13    }
```

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1  class MyTests : BehaviorSpec({
2      given("a broomstick") {
3          `when`("I sit on it") {
4              then("I should be able to fly") {
5                  // test code
6              }
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Another flavor of Kotest

Kotest: <https://kotest.io>

```
1  class MyTests : DescribeSpec({
2      describe("score") {
3          it("start as zero") { /* test here */ }
4          describe("with a strike") {
5              it("adds ten") { /* test here */ }
6              it("carries strike to the next frame") { /* test here */ }
7          }
8          describe("for the opposite team") {
9              it("Should negate one score") { /* test here */ }
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Migrating from JUnit?

```
1  class AnnotationSpecExample : AnnotationSpec() {  
2      @BeforeEach  
3      fun beforeTest() {  
4          println("Before each test")  
5      }  
6      @Test  
7      fun test1() {  
8          1 shouldBe 1  
9      }  
10     @Test  
11     fun test2() {  
12         3 shouldBe 3  
13     }  
14 }
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explanation shouldBe better

Kotest has its own assertions

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- kotest-assertions-core

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- kotest-assertions-core
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explanation shouldBe better

Kotest has its own assertions

- kotest-assertions-core
- kotest-assertions-json
- kotlintest-assertions-jsoup
- And many others

Core Assertions

```
1 1 shouldBe 1
2 1 shouldNotBe 2
3 "Hello" shouldBe "Hello"
4 listOf(1,2,3) shouldContain 2
5 mapOf("a" to 1, "b" to 2) shouldContainKey "a"
6 null shouldBe null
7 "abc" shouldStartWith "a"
8 "xyz" shouldEndWith "z"
9 5 shouldBeGreaterThan 3
10 2 shouldBeLessThan 5
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JSON Assertions

```
1  "{}".shouldBeValidJson()
2  "{}".shouldBeJsonObject()
3  a.shouldEqualJson(b, compareJsonOptions { arrayOrder = ArrayOrder.Strict })
```

Or even...

JSON Assertions

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1  "{}".shouldBeValidJson()  
2  "{}".shouldBeJsonObject()  
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```

Or even...

Schema Validation

```
1  val addressSchema = jsonSchema {
2    obj { // object is reserved, obj was chosen over jsonObject for brevity but could be changed
3      withProperty("street", required = true) { string() }
4      withProperty("zipCode", required = true) {
5        integer {
6          beEven() and beInRange(10000..99999) // supports constructing a matcher that will
7        }
8      }
9      additionalProperties = false // triggers failure if other properties are defined in an object
10   }
11 }
12
13 val personSchema = jsonSchema {
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15     withProperty("name", required = true) { string() }
16     withProperty("address") { addressSchema() } // Schemas can re-use other schemas 🎉
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Schema Validation

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22 // fails with:
23 // $.name => Expected string, but was undefined
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25 "" { "name": "Emil", "age": 34 } ""
26 // Passes, since address isn't required and `additionalProperties` are allowed
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Jsoup

```
1 element.shouldHaveChildWithTag(tag)
2 element.shouldHaveText(text)
3 element.shouldHaveAttribute(name)
4 element.shouldHaveSrc(src)
```

50 shades of kotest

<https://kotest.io>

Fun spec String Spec

Should spec Describe spec

Behaviour spec Word spec

Free spec Feature spec

Expect spec Annotation spec

layout: center

How do I run the same tests with different inputs?

Data-driven testing a.k.a Parametrized tests

Parametrized tests? Java

```
1  static Stream<Arguments> stringIntAndListProvider() {  
2      return Stream.of(  
3          arguments("apple", 1, Arrays.asList("a", "b")),  
4          arguments("lemon", 2, Arrays.asList("x", "y"))  
5      );  
6  }  
7  @ParameterizedTest  
8  @MethodSource("stringIntAndListProvider")
```

Parametrized tests? Java

```
4         arguments("lemon", 2, Arrays.asList("x", "y"))
5     );
6 }
7 @ParameterizedTest
8 @MethodSource("stringIntAndListProvider")
9 void testWithMultiArgMethodSource(String str, int num, List<String> list) {/
10     // snip
11 }
```

Parametrized tests? Java

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`stringIntAndListProvider`, really?

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`stringIntAndListProvider`, really?

- Becomes cluttered
- Not typed (enough)

Parametrized tests? Java

```
1  static Stream<Arguments> stringIntAndListProvider() {  
2      return Stream.of(  
3          arguments("apple", 1, Arrays.asList("a", "b")),  
4          arguments("lemon", 2, Arrays.asList("x", "y"))  
5      );  
6  }  
7  @ParameterizedTest  
8  @MethodSource("stringIntAndListProvider")
```

`stringIntAndListProvider`, really?

- Becomes cluttered
- Not typed (enough)
- Bound by a string!

Parametrized tests? Kotlin (with kotest)

```
1  context("Pythag triples tests") {  
2      withData(  
3          PythagTriple(3, 4, 5),  
4          PythagTriple(6, 8, 10),  
5          PythagTriple(8, 15, 17),  
6          PythagTriple(7, 24, 25)  
7      ) { (a, b, c) ->  
8          isPythagTriple(a, b, c) shouldBe true  
9      }  
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7      ) { (a, b, c) ->  
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9      }  
10 }
```

layout: image image: /confused.png

What if

I need to test things I don't know I need to
test?

clicks: 5

Never write your own hashing (unless...)

```
1  withData(  
2    "love",  
3    "secret",  
4    "god", // Who will remember the reference?  
5  ) { input ->  
6    MD5.compute(input) shouldBe libMd5(input)  
7  }
```

clicks: 5

Never write your own hashing (unless...)

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1  withData(  
2    "love",  
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Launching...

clicks: 5

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```

Launching...

1. ✓ love
2. ✓ secret
3. ✓ god

layout: image image: /lens.png

Time to investigate!

Property tests!

To test things we can not even *imagine*

layout: statement

Demo

Another example

```
1  checkAll(  
2    Arb.string(  
3      codepoints = Arb.of(  
4        (Char.MIN_VALUE.code..Char.MAX_VALUE.code).map(::Codepoint)  
5      ),  
6      range = 0..52  
7    )  
8  ) { input ->  
9    MD5.compute(input) shouldBe libMd5(input)  
10 }
```

Another example

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1  checkAll(  
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3      codepoints = Arb.of(  
4        (Char.MIN_VALUE.code..Char.MAX_VALUE.code).map(::Codepoint)  
5      ),  
6      range = 0..52  
7    )  
8  ) { input ->  
9    MD5.compute(input) shouldBe libMd5(input)  
10 }
```

layout: image image: /explode.png

Error!

[illegible]

Wow, it was a complex sample!

But there is more...

Error!

Wow, it was a complex sample!

But there is more...

Error!

Wow, it was a complex sample!

But there is more...

Error!

[illegible]

Wow, it was a complex sample!

But there is more...

Error!

[illegible]

And there is more!

Error!

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Error!

```
1 Property failed after 1 attempts
```

```
3 Arg 0: "☒" (shrunk from ☐ℓooooooooooooooooooooooooooooÅoooooŌoooooooooňoooo)
```

```
5 Repeat this test by using seed 4919313423746630853
```

Error!

```
1  Property failed after 1 attempts
2
3  Arg 0: "☒" (shrunk from ☐лoooooooooooooooooooooÃoooooŎoooooooooňoooo)
4
5  Repeat this test by using seed 4919313423746630853
```

Error!

```
3 Arg 0: "☒" (shrunk from ☐ℓooooooooooooooooooooooooooooÅoooooŌoooooooooňoooo)
```

Error!

```
5 Repeat this test by using seed 4919313423746630853
```

Repeat? Yes!

```
1  checkAll(  
2    PropTestConfig(seed = 4919313423746630853),  
3    Arb.string(  
4      codepoints = Arb.of(  
5        ('a'.code..Char.MAX_HIGH_SURROGATE.code)  
6        .map(::Codepoint)  
7      ),  
8    )  
9  ) { input ->  
10    MD5.compute(input) shouldBe libMd5(input)  
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```

- reproduce the test **exactly** like in the failed instance
- debug it
- fix it
- PROFIT!!

Repeat? Yes!

```
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- reproduce the test **exactly** like in the failed instance
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layout: image image: /mockingbird.png

Time to mock!

Disclaimer

I believe one should mock, not fake. Because

1. Granular control
2. Can check that something was NOT called

Time to mock

Setup

```
1  class UserService(val repo: UserRepository) {  
2      fun getAllUsers() = repo.allUsers()  
3  }  
4  
5  class UserRepository {  
6      fun allUsers(): List<User> { TODO("demo") }  
7  }  
8  
9  class User(val name: String, val age: Int)
```

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Time to mock

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```

How the test looks in Java

```
1  UserRepository mockRepository = Mockito.mock(UserRepository.class);
2
3  // Create some sample user data
4  List<User> users = Arrays.asList(new User("Alice", 25), new User("Bob", 30), new User("Charlie", 35));
5
6  // Define the expected behavior of the mock repository
7  Mockito.when(mockRepository.allUsers()).thenReturn(users);
8
9  // Create an instance of the UserService with the mock repository
10 UserService userService = new UserService(mockRepository);
11
12 // Call the method under test
13 List<User> result = userService.getAllUsers();
14
15 // Verify the result
16 Assertions.assertEquals(users, result);
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Time to mock! (MockK edition)

```
1  val users = listOf(User("Alice", 25), User("Bob", 30), User("Charlie", 35))
2  val repo = mockk<UserRepository> {
3      every { allUsers() } returns users
4  }
5  val service = UserService(repo)
6  service.getAllUsers() shouldBe users
```

Time to mock! (MockK edition)

```
1  val users = listOf(User("Alice", 25), User("Bob", 30), User("Charlie", 35))
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5  val service = UserService(repo)
6  service.getAllUsers() shouldBe users
```

Nice DSL

Code describing `UserRepository` is inside `{}`

What if I want to mock users too?

Time to mock users! (mockk edition)

```
1  val repo = mockk<UserRepository> {
2      every { allUsers() } returns listOf(
3          mockk {
4              every { name } returns "Pasha"
5          },
6          mockk {
7              every { name } returns "Alexandra"
8          },
9      )
10 }
11 val service = UserService(repo)
12 service.getAllUsers()[1].name shouldBe "Alexandra"
```

Time to mock users! (mockk edition)

```
1  val repo = mockk<UserRepository> {  
2      every { allUsers() } returns listOf(  
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4              every { name } returns "Pasha"  
5          },  
6          mockk {  
7              every { name } returns "Alexandra"  
8          },  
9      )  
10 }  
11 val service = UserService(repo)  
12 service.getAllUsers()[1].name shouldBe "Alexandra"
```

Time to mock users! (mockk edition)

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7              every { name } returns "Alexandra"  
8          },  
9      )  
10 }  
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12 service.getAllUsers()[1].name shouldBe "Alexandra"
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Time to mock users! (mockk edition)

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7              every { name } returns "Alexandra"  
8          },  
9      )  
10 }  
11 val service = UserService(repo)  
12 service.getAllUsers()[1].name shouldBe "Alexandra"
```

Users are described inside `User` mocks!

Atrium expectations!

<https://docs.atriumlib.org/>

```
1  expect(service.getAllUsers()) toContain o inAny order but only the entries(  
2    {  
3      its { name } toEqual "Pash"  
4      its { age } toEqual 39  
5    },  
6    {  
7      its { name } toEqual "Mark"  
8      its { age } toEqual 16  
9    }  
10  )
```

Atrium expectations!

<https://docs.atriumlib.org/>

```
1  expect(service.getAllUsers()) toContain o inAny order but only the entries(  
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Atrium expectations!

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9    }  
10  )
```

I've made a mistake to show the error message

Atrium expectations!

<https://docs.atriumlib.org/>

```
1  expect(service.getAllUsers()) toContain o inAny order but only the entries(  
2    {  
3      its { name } toEqual "Pash"  
4      its { age } toEqual 39  
5    },  
6    {  
7      its { name } toEqual "Mark"  
8      its { age } toEqual 16  
9    }  
10  )
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I've made a mistake to show the error message

Error report

```
1 I expected subject: [User(#3), User(#4)] (java.util.Arrays.ArrayList <315506631>)
2 ♦ to contain only, in any order:
3   ✕ an element which needs:
4     » ► its.definedIn(MockTest.kt:42):
5       ■ to equal: "Pash" <2026086646>
6     » ► its.definedIn(MockTest.kt:43):
7       ■ to equal: 36 (kotlin.Int <116289638>)
8   ✓ an element which needs:
9     » ► its.definedIn(MockTest.kt:46):
10      ■ to equal: "Mark" <574921197>
11     » ► its.definedIn(MockTest.kt:47):
12      ■ to equal: 13 (kotlin.Int <453296736>)
13   !! following elements were mismatched:
14     ◦ User(#3) (User <1041341319>)
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- Readable as a text
- Shows the errors

Dessert

Now, when we saw different nice DSL-using capabilities...

```
1  ideaTest<App>("Create Kafka connection") {
2      showBDTPanel {
3          createBaseConnection("Kafka") {
4              fillCredentialsForKafka(connectionName)
5              checkSuccessfulMessage()
6              ok()
7          }
8          checkMonitoringConnection(monitoredToolWindow)
9      }
10 }
```

layout: image image: /magick.png

How does it work? Magick!

Dive deeper

layout: image image: /question.png

What did we learn?

What did we learn?

1. Tests should be readable
2. Testing is not "just unit tests" sometimes
3. Kotest gives good structure to the tests.
4. Kotest has built-in property tests
5. Atrium gives us assertions DSL

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Thank you! Time for questions 😊

 asm0di0

 me@asm0dey.site

 @asm0dey.site

    asm0dey

 <https://linktr.ee/asm0dey>

layout: end

Deeper dive into JetBrains tests

```
1  ideaTest<App>("Create Kafka connection") {
```

really means:

```
1  fun <T : App> Group.ideaTest(title: String,  
2                                enabled: Boolean = true,  
3                                isMuted: Boolean = false,  
4                                testBody: T.(ctx: IdeaNodeContext<T>) -> Unit) {  
5      IdeaNodeContext<T>(scenarioContext).apply {  
6          test(title, enabled, isMuted) {  
7              app.testBody(this)  
8          }  
9      }  
10 }
```

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Deeper dive into JetBrains tests

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1      showBDTPanel {
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really means:

```
1  fun App.showBDTPanel(block: BDTPanel.() -> Unit) {
2      val app = this
3      ideaFrame {
4          val frame = this
5          step("Open Big Data Tools Panel") {
6              openBDTPanel(app)
7              if (isKafka()) {
8                  try {
9                      container(byXPath("//div[@accessiblename='Kafka Tool Window' and @class=
10                          BDTPanel(app, frame, this).block()
11                      })
```

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Deeper dive into JetBrains tests

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1      createBaseConnection("Kafka") {
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really means:

```
1  fun createBaseConnection(typeOfConnection: String, block: ConnectionDialog.() -> Unit) {  
2      step("Open connection dialog") {  
3          if (app.isKafka()) {  
4              frame.find<ComponentFixture>(byXPath("//div[contains(@tooltiptext.key, 'toolwin  
5                  clickTypeOfConnection(typeOfConnection, block)  
6                  pause(5000)
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layout: center

And this is the abyss!

But you can do the same!

[Go back to summary](#)

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